Assessment Report

Assessment Name: Olympic Database

Student Name : Ming Cheing Yee

Curtin Student ID : 22467577

Lab Group : 2 (Friday, 3pm-5pm)

Introduction

This report aims to provide an explanation of the developing and implementing of the database called Olympic_22467577. The selected scenario of this database is the Olympic Games 2024, which took place from 26 July to 11 August 2024, in Paris, France. The users can store and retrieve data, including athletes, officials, events, venues, countries, medals, participation, and event management, from the database for any academic purpose. As a result, it will serve as a valuable resource for any researchers, students, or even future Olympic Games organizers to make analysis and interpretations on data that is related to the Olympics.

During the development of the database, I have done many activities, including identifying entities, constraints, and relationships in the database's structure, designing an entity-relationship (ER) diagram and also relational schema to ensure their normality, and lastly, searching for valid resources about the Olympic Games 2024 data. Furthermore, I use MySQL to implement the database on Linux and insert pieces of valid data into the database.

This report will show all the details of the designing process of the database, including the reasons for the selection of entities and attribute’s data types, the implementation process, various queries, and the advanced features developed, which are the stored procedures and triggers that were used to increase the efficiency of the uses of the database. The aim of this report is to provide a clear insight into the database's functionality and its significance in supporting Olympic-related data management.

Database Designing

I. Entities selected

Entity Sets	Primary Keys	Other Attribute	Reason of selection
Athletes	Athlete_Code	Name, Gender, Birth_Date, Discipline, Side_Job	The main participant of the Olympic
Countries	Country_Code	Name	The land that athletes belong to
Events	Sport_Code	Name	Each competition has its own event
Venues	Venue_Code	Name	The location where events take place during competition
Officials	Official_Code, Belonged Organization	Name, Gender, Role	People play a role to oversee the events
Medals (Weak Entity)	Medal_Code (Weak Key)	Medal_Type	The result of the events

II. Relationship selected

Relationship Sets	Related Entity Sets	Other Attribute	Reason of selection
Participate	Athletes & Events	Started_Date, Ended_Date	Athletes participate events
Held in	Events & Venues		Events held in venues
Managed by	Events & Officials		Events managed by officials
Belongs to	Athletes & Countries		Athletes belonged to country
Wins (Weak Relationship)	Athletes & Medals	Awarded_Date	Athletes win medal

III. Relationship constraints

Relationship Sets	Cardinality Constraints	Participation Constraints
Participate	Many to Many ➔ One athlete can participate many events ➔ One event can be participated by many athlete	**Athlete ->Total** (Athlete cannot exist without event) **Event -> Partial** (Event can exist without athlete)
Held in	One to Many ➔ One event can hold in only one venue ➔ One venue can hold many events	**Event ->Partial** (Event can exist without venue) **Venue->Partial** (Venue can exist without event)
Managed by	Many to Many ➔ One event can be managed by many officials ➔ One official can manage many event	**Event->Partial** (Event can exist without official) **Official->Total** (Official cannot exist without event)
Belongs to	One to Many ➔ One athlete belongs to one country only ➔ One country can have many athletes	**Athlete->Total** (Athlete cannot exist without country) **Country->Partial** (Country can exist without athlete)
Wins (Weak Relationship)	One to Many ➔ One athlete can award many medals ➔ One medal can only be awarded by one athlete	**Athlete->Total** (Athlete cannot exist without medal) **Medal->Partial** (Medal can exist without athlete)

IV. ER Model

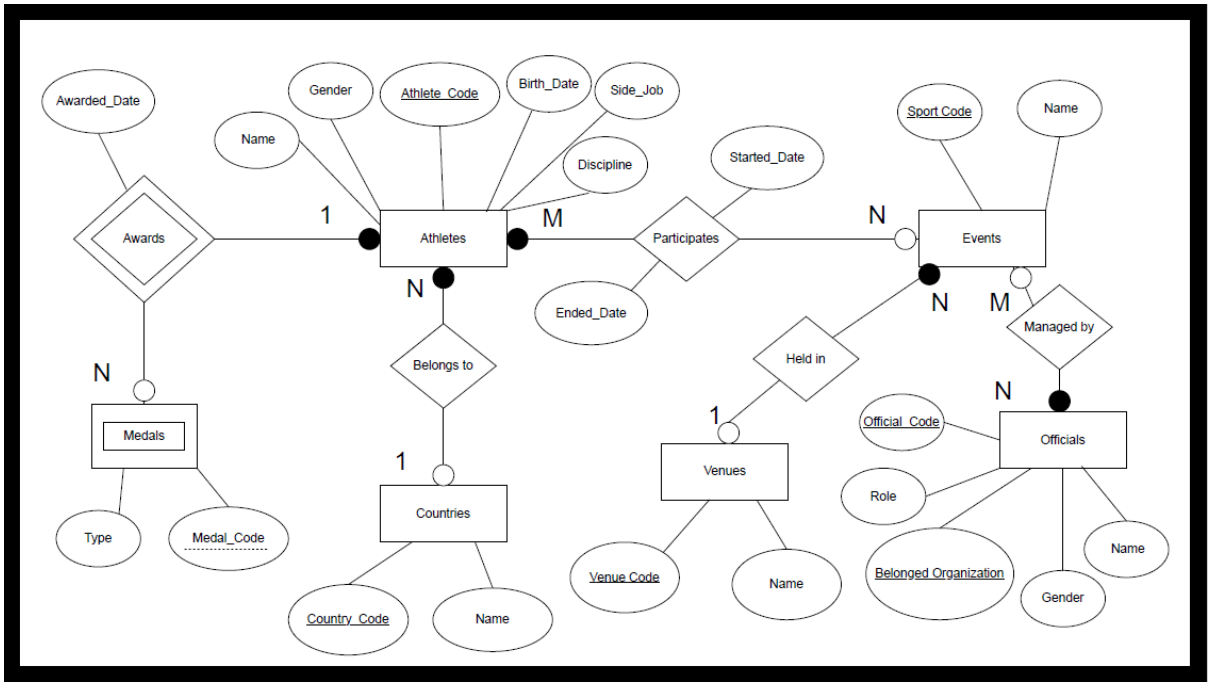


Figure 1- ER MODEL

This ER model is produced using Chen notation, which visualizes the database structure as a reference during the implementation and helps future designers make changes or updates to my database if needed

V. Relational schema

Venue (Venue_Code , Name)

Country (Country_Code, Name)

Athlete (Athlete_Code, Name, Gender, Discipline, Birth_Date, Side_Job, Country_Code)

FK Country_Code REF Country (Country_Code)

Event (Sport_Code, Name, Venue_Code)

FK Venue_Code REF Venue (Venue_Code)

Medal (Athlete_Code, Medal_Code, Type)

FK Athlete_Code REF Athlete (Athlete_Code)

Participates (Athlete_Code, Sport_Code, Started_Date, Ended_Date)

FK Athlete_Code REF Athlete (Athlete_Code)

FK Sport_Code REF Event (Sport_Code)

Official (Official_Code, Belonged Organization, Name, Gender, Role)

Managed by (Official_Code, Belonged_Organization, Sport_Code)

FK Official_Code REF Official (Official_Code)

FK Belonged_Organization REF Official (Belonged_Organization)

FK Sport_Code REF Event (Sport_Code)

Normality Check

i. In First Normal Form
   ➔ No multivalued attribute in every entity
ii. In Second Normal Form
   ➔ Already in first normal form
   ➔ Partial dependency eliminated
   ➔ Non-key attribute only depends on primary key
iii. In Third Normal Form
   ➔ Already in second normal form
   ➔ No non-key attribute depends on another non-key attribute

VI. Data Description

Venue (Venue Table)

COLUMN NAME	DATA TYPE	SIZE	Constraints	DESCRIPTION
Venue_Code	CHAR	3	PRIMARY KEY	Identify Code for Each Venue used
Name	VARCHAR	50	NOT NULL	Name of Each Venue used

Country (Country Table)

COLUMN NAME	DATA TYPE	SIZE	Constraints	DESCRIPTION
Country_Code	CHAR	3	PRIMARY KEY	Identify Code for Each Country Joined
Name	VARCHAR	40	NOT NULL	Name of Each Country Joined

Athlete (Athlete Table)

COLUMN NAME	DATA TYPE	SIZE	Constraints	DESCRIPTION
Athlete_Code	INT	7	PRIMARY KEY	Identify Code for each athlete participated
Name	VARCHAR	50	NOT NULL	Name of each athlete
Gender	VARCHAR	6	NOT NULL	Gender for each athlete (Male/Female)
Discipline	VARCHAR	50		The sport an athlete focusses on and compete
Birth_Date	DATE		CHECK (Birth_Date >= '1900-01-01')	Date of Born (Athlete)
Side_Job	VARCHAR	50		The other job an athlete does besides the athlete
Country_Code	CHAR	3	NO NULL FOREIGN KEY	Identify the nationality of Athlete

Medal (Medal Table)

COLUMN NAME	DATA TYPE	SIZE	Constraints	DESCRIPTION
Athlete_Code	INT	7	FOREIGN KEY PRIMARY KEY	Identify Code of Athlete get the medal
Medal_Code	INT	1	PRIMARY KEY	Identify Code of Medal (1 for Gold,2 for Silver,3 for Bronze)
Medal_Type	CHAR	6	NOT NULL	The material used to make medal
Award_Date	DATE			The date an athlete gets the medal

Event (Event Table)

COLUMN NAME	DATA TYPE	SIZE	Constraints	DESCRIPTION
Sport_Code	CHAR	3	PRIMARY KEY	Identify Code of sport type
Name	VARCHAR	50	NOT NULL	Name of sport type competed
Venue_Code	CHAR	3	FOREIGN KEY NOT NULL	Identify Code of Venue used to hold an event (All the event of same sport type holds in the same venue)

Participation (Participates Relationship Table)

COLUMN NAME	DATA TYPE	SIZE	Constraints	DESCRIPTION
Athlete_Code	INT	7	FOREIGN KEY NOT NULL	Identify Code of Athlete participate in an event
Sport_Code	CHAR	3	FOREIGN KEY NOT NULL	Identify Code of Event an athlete has participated
Started_Date	DATE			The date of an event started
Ended_Date	DATE			The date of an event ended

Official (Officials Table)

COLUMN NAME	DATA TYPE	SIZE	Constraints	DESCRIPTION
Official_Code	INT	7	PRIMARY KEY	Identify Code of Official
Name	VARCHAR	50	NOT NULL	Name of each Official
Gender	CHAR	6		Gender of each Official (Male/Female)
Role	VARCHAR	30		Function of official in an event
Belonged_Organization	VARCHAR	80	PRIMARY KEY	Name of Organization of an Official belonged

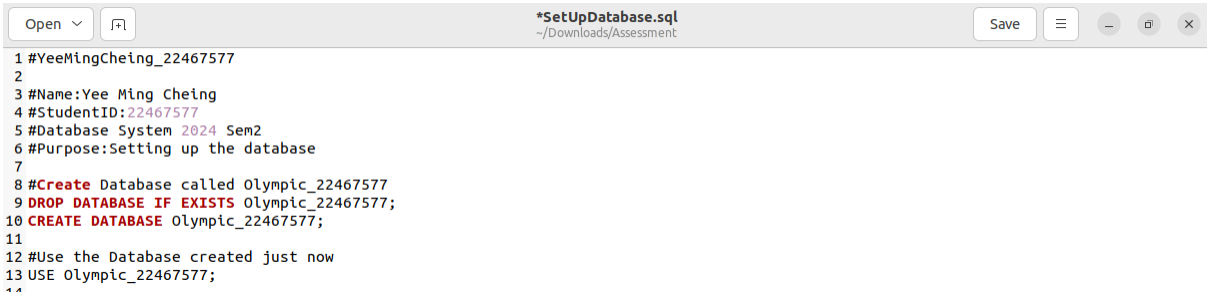
Event_Management (Manages relationship Table)

COLUMN NAME	DATA TYPE	SIZE	Constraints	DESCRIPTION
Official_Code	int	7	FOREIGN KEY NOT NULL	Identify Code of Official that manage an event
Belonged_Organization	VARCHAR (80)	50	FOREIGN KEY NOT NULL	Name of Organization of an Official (manage an event) belonged
Sport_Code	CHAR	3	FOREIGN KEY NOT NULL	Identify Code of Sport of event that an official manages

Implementation

In order to implement the database, I create four SQL script:

1) SetUpDatabase.sql
   ➔Consists the commands that create Olympic_22467577 database




```
#YeeMingCheing_22467577

#Name:Yee Ming Cheing
#StudentID:22467577
#Database System 2024 Sem2
#Purpose:Setting up the database

#Create Database called Olympic_22467577
DROP DATABASE IF EXISTS Olympic_22467577;
CREATE DATABASE Olympic_22467577;

#Use the Database created just now
USE Olympic_22467577;
```

Figure 2-Section of command Details of SetUpDatabase.sql

2) SetUpDataTable.sql
   ➔Consists the commands that create all the related table into the Olympic_22467577 database


```
#YeeMingCheing_22467577

#Name:Yee Ming Cheing
#StudentID:22467577
#Database System 2024 Sem2
#Purpose:Setting up the datatable

#Create the Country Table
DROP Table IF EXISTS Country;
CREATE table Country(
    Country_Code CHAR(3) PRIMARY KEY, -- Set Country Code As Primary Key
    Name VARCHAR(40) NOT NULL
);

#Create the Venue Table
DROP Table IF EXISTS Venue;
CREATE table Venue(
    Venue_Code CHAR(3) PRIMARY KEY, -- Set Venue Code As Primary Key
    Name VARCHAR(50) NOT NULL
);

#Create the Athlete Table
DROP Table IF EXISTS Athlete;
CREATE table Athlete(
    Athlete_Code Int(7) PRIMARY KEY, -- Set Athlete Code As Primary Key
    Name VARCHAR(50) NOT NULL,
    Gender VARCHAR(6) NOT NULL,
    Discipline VARCHAR(50),
    Birth_Date DATE CHECK (Birth_Date >= '1900-01-01'), -- Use CHECK Constraint to ensure a valid athlete's age
    Side_Job VARCHAR(50) DEFAULT 'None', -- Set the Default Value of Side_Job is 'None' for the athlete who does not have side job
    Country_Code CHAR(3) NOT NULL,
    CONSTRAINT fk_country
        FOREIGN KEY (Country_Code)
        REFERENCES Country(Country_Code) -- Country Code from Country table become foreign key
        ON DELETE CASCADE ON UPDATE CASCADE -- Delete or Update on Country Code in Country Table will change the Country Code in Athlete Table
);

#Create the Medal Table
```

Figure 3-Section of command Details of SetUpDataTable.sql

3) SetUpData.sql

➔ Consists the commands that insert data into all the table in the Olympic_22467577 database

➔ Data source: https://www.kaggle.com/datasets/piterfm/paris-2024-olympic-summer-games

➔ I) Country table -> 224 rows of data

II) Venue table -> 34 rows of data

III) Athlete table -> 400 rows of data

IV) Medal table -> 10 rows of data

V) Event table -> 34 rows of data

VI) Participation table -> 400 rows of data

VII) Official table -> 545 rows of data

VIII) Event_Management table -> 545 rows of data

Figure 4-Section of command Details of SetUpData.sql

4) SettingUp.sql

➔ Consists of the commands that call all the SQL scripts related to database implementation, such as SetUpDatabase.sql or SetUpDataTable.sql

➔ The user only needs to call this SQL script to complete the implementation on the Olympic_22467577 database

Open *SettingUp.sql ~/Downloads/Assessment Save

```
#YeeMingCheing_22467577

#Name:Yee Ming Cheing
#StudentID:22467577
#Database System 2024 Sem2
#Purpose:Call the script that able to Setup the database,table,and insert data into the data

#Call the script that set the database user
source Background.sql;

#Call the script that Create the Database
source SetUpDatabase.sql;

#Call the script that Create the Table
source SetUpDataTable.sql;

#Call the script that insert the data into the Table
source SetUpData.sql;

#Call the script that create the stored procedure
source StoredProcedure.sql;

#Call the script that create the Triggers
source Trigger.sql;

#Call the script that create the Views
source View.sql;

```

Figure 5-Command Details of SettingUp.sql

yee@yee-virtual-machine: ~/Downloads/Assessment

```
mysql> source SettingUp.sql;
Query OK, 0 rows affected (0.04 sec)

Query OK, 0 rows affected (0.02 sec)

Query OK, 0 rows affected (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

Query OK, 10 rows affected (0.18 sec)

Query OK, 1 row affected (0.01 sec)

Database changed
Query OK, 0 rows affected, 1 warning (0.02 sec)

Query OK, 0 rows affected (0.07 sec)

Query OK, 0 rows affected, 1 warning (0.00 sec)

Query OK, 0 rows affected (0.06 sec)

Query OK, 0 rows affected, 1 warning (0.01 sec)

Query OK, 0 rows affected, 1 warning (0.06 sec)

Query OK, 0 rows affected, 1 warning (0.00 sec)

Query OK, 0 rows affected, 2 warnings (0.06 sec)

Query OK, 0 rows affected, 1 warning (0.00 sec)

Query OK, 0 rows affected (0.07 sec)

Query OK, 0 rows affected, 1 warning (0.01 sec)

Query OK, 0 rows affected, 1 warning (0.07 sec)

Query OK, 0 rows affected, 1 warning (0.01 sec)
```

Figure 6-Evidence of Implementation using SettingUp.sql

Use of Database

I.Querying

➔8 questions are prepared for the query implementation

➔All the queries are stored in the Querying.sql script

1. Which athlete has a side job and is aged between 19 and 29?

   ➔ Shows the percentage of athlete that able to balance sport and career after graduated from high school

```
SELECT Athlete_Code, Name, Side_Job

FROM Athlete

WHERE Side_Job IS NOT NULL AND

Birth_Date BETWEEN '1995-01-01' AND '2005-12-31'
```

Figure 7-Query for Question1

```
mysql> SELECT Athlete_Code, Name, Side_Job
    -> FROM Athlete
    -> WHERE Side_Job IS NOT NULL AND
    -> Birth_Date BETWEEN '1995-01-01' AND '2005-12-31'
    -> ;
+--------------+----------------------------+-------------------------------------------+
| Athlete_Code | Name                       | Side_Job                                  |
+--------------+----------------------------+-------------------------------------------+
|      1533136 | BASS BITTAYE Gina Mariam   | police officer (sub-inspector)            |
|      1533176 | CAMARA Ebrahima            | prison officer                            |
|      1533235 | ARELLANO Fernanda          | student                                   |
|      1533237 | TOSCANO Pamela             | sport soldier                             |
|      1533364 | ISMAIL Mohamed             | sport soldier                             |
|      1533371 | HASSAN Ibrahim             | soldier with the Djiboutian Republican Guard |
|      1533372 | WAISS Abdi                 | sport soldier                             |
|      1533374 | HASSAN NOUR Samiyah        | sport soldier                             |
|      1535191 | GONZALEZ Alegna            | serves in the navy                        |
|      1535226 | AGUILAR CHAVEZ PEON Mariana | student                                   |
|      1535310 | BLANCO Luisa               | student                                   |
|      1535343 | WILLARS VALDEZ Randal      | Armed forces                              |
|      1535346 | OROZCO LOZA Alejandra      | Armed forces                              |
|      1535348 | AGUNDEZ GARCIA Gabriela    | Armed forces                              |
|      1535364 | CARRILLO Duilio            | Armed Forces                              |
|      1535367 | SOUZA Daniela Paola        | Military                                  |
|      1535385 | HOUSSEIN Alexandre         | student                                   |
|      1535395 | AL SAYEGH Safia            | student                                   |
|      1535430 | GRANDE Matias              | student                                   |
|      1535438 | RAMIREZ RAMOS Edson Ismael | student                                   |
|      1535441 | QUEZADA MARQUEZ Carlos     | student                                   |
|      1535505 | LINARES Natalia            | student                                   |
```

Sample Answer for Question 1

2. Which official comes from a sports federation?

➔ Shows the percentage of official from a sport federation

```
SELECT Official_Code, Name, Role

FROM Official

WHERE Belonged_Organization LIKE '%Federation%';
```

Figure 8-Query for question2

```
mysql> SELECT Official_Code, Name, Role
    -> FROM Official
    -> WHERE Belonged_Organization LIKE '%Federation%';
+---------------+----------------------------------+-------------+
| Official_Code | Name                             | Role        |
+---------------+----------------------------------+-------------+
|       1550642 | FUMEA Maria                      | Judge       |
|       1550643 | LEUNG Li Li                      | Judge       |
|       1550645 | MIAO Zhongyi                     | Judge       |
|       1550646 | VALENZO AOKI Naomi Chieko        | Judge       |
|       1550647 | AL HITMI Ali                     | Judge       |
|       1550648 | AL TABBAA Mouhammed Youssef      | Judge       |
|       1550649 | CELEN Suat                       | Judge       |
|       1550650 | ESAWY Ehab                       | Judge       |
|       1550651 | YAGI KITAGAWA Tammy              | Judge       |
|       1550652 | VALERIO Rogerio                  | Judge       |
|       1550653 | MOWBRAY Denis                    | Judge       |
|       1550654 | PEGAN Aljaz                      | Judge       |
|       1550655 | PENICHE FRANCO Alejandro         | Judge       |
|       1550656 | PERRIERE Charles                 | Judge       |
|       1550657 | TAEYMANS Rosy                    | Judge       |
|       1571122 | ALSHATHRI A Rahman               | Judge       |
|       1571123 | KIM Nellia                       | Judge       |
|       1571124 | TITOV Vasilii                    | Judge       |
|       1893012 | CHIARI Roberto                   | Referee     |
|       1893016 | THOMSON Mike                     | Referee     |
|       1893040 | DUPUCHE Crystal                  | Referee     |
```

Sample Answer for Question 2

3. How many athletes in the Olympics does a country have?
   ➔ The data obtained can be used in analysis the overall performance of a country

```
SELECT Country.Name, Country.Country_Code, COUNT(Athlete.Country_Code)
AS 'Number of athlete represented'
FROM Country INNER JOIN Athlete ON Country.Country_Code = Athlete.Country_Code
Group BY Country.Country_Code , Country.Name
ORDER BY COUNT(Athlete.Country_Code) DESC;
```

Figure 9-Query for Question 3

```
mysql> SELECT Country.Name, Country.Country_Code, COUNT(Athlete.Country_Code)
    -> AS 'Number of athlete represented'
    -> FROM Country INNER JOIN Athlete ON Country.Country_Code = Athlete.Country_Code
    -> Group BY Country.Country_Code , Country.Name
    -> ORDER BY COUNT(Athlete.Country_Code) DESC;
+------------------------------+--------------+------------------------------+
| Name                         | Country_Code | Number of athlete represented |
+------------------------------+--------------+------------------------------+
| Mexico                       | MEX          |                           83 |
| Argentina                    | ARG          |                           58 |
| India                        | IND          |                           42 |
| Ireland                      | IRL          |                           26 |
| Islamic Republic of Iran     | IRI          |                           25 |
| Ethiopia                     | ETH          |                           20 |
| Iraq                         | IRQ          |                           16 |
| Singapore                    | SGP          |                           13 |
| Romania                      | ROU          |                           13 |
| Colombia                     | COL          |                           10 |
| Puerto Rico                  | PUR          |                            9 |
| United Arab Emirates         | UAE          |                            7 |
| Djibouti                     | DJI          |                            7 |
| Jamaica                      | JAM          |                            6 |
| Armenia                      | ARM          |                            5 |
| Guyana                       | GUY          |                            5 |
| Saint Lucia                  | LCA          |                            4 |
| Malaysia                     | MAS          |                            4 |
| Côte d'Ivoire                | CIV          |                            4 |
| AIN                          | AIN          |                            4 |
| Comoros                      | COM          |                            3 |
| Congo                        | CGO          |                            3 |
| Jordan                       | JOR          |                            3 |
| Marshall Islands             | MHL          |                            3 |
| Uzbekistan                   | UZB          |                            3 |
| Oman                         | OMA          |                            2 |
| Saint Kitts and Nevis        | SKN          |                            2 |
| Malawi                       | MAW          |                            2 |
| St Vincent and the Grenadines | VIN          |                            2 |
| Libya                        | LBA          |                            2 |
```

Sample Answer for Question 3

4. How many competitions have been held in a venue?
   ➔ The data obtained can be used for resource allocation in future Olympic events

```
SELECT Venue.Venue_Code, Venue.Name, COUNT(Event.Venue_Code) AS 'Total number of uses'

FROM Venue LEFT OUTER Join Event ON Venue.Venue_Code = Event.Venue_Code

GROUP BY Venue.Venue_Code,Venue.Name

ORDER BY COUNT(Event.Venue_Code) DESC;
```

Figure 10-Query for Question 4

```
mysql> SELECT Venue.Venue_Code, Venue.Name, COUNT(Event.Venue_Code) AS 'Total number of uses'
    -> FROM Venue LEFT OUTER Join Event ON Venue.Venue_Code = Event.Venue_Code
    -> GROUP BY Venue.Venue_Code,Venue.Name
    -> ORDER BY COUNT(Event.Venue_Code) DESC;
+------------+--------------------------------------+----------------------+
| Venue_Code | Name                                 | Total number of uses |
+------------+--------------------------------------+----------------------+
| SP         | South Paris Arena                    |                    4 |
| VN         | Vaires-sur-Marne Nautical Stadium    |                    3 |
| BCY        | Bercy Arena                          |                    2 |
| CDM        | Champ de Mars Arena                  |                    2 |
| GRP        | Grand Palais                         |                    2 |
| STA        | Stade de France                      |                    2 |
| LC         | La Concorde                          |                    2 |
| RGA        | Stade Roland-Garros                  |                    2 |
| MAM        | Marseille Marina                     |                    1 |
| YDM        | Yves-du-Manoir Stadium               |                    1 |
| VER        | Château de Versailles                |                    1 |
| VE2        | Saint-Quentin-en-Yvelines BMX Stadium |                    1 |
| VE1        | Saint-Quentin-en-Yvelines Velodrome  |                    1 |
| TRO        | Trocadéro                            |                    1 |
| TAH        | Teahupo'o, Tahiti                    |                    1 |
| NPA        | North Paris Arena                    |                    1 |
| AQC        | Aquatics Centre                      |                    1 |
| BOR        | Bordeaux Stadium                     |                    1 |
| CPL        | Porte de La Chapelle Arena           |                    1 |
| CTX        | Chateauroux Shooting Centre          |                    1 |
| DEF        | Paris La Defense Arena               |                    1 |
| EIF        | Eiffel Tower Stadium                 |                    1 |
| LBO        | Le Bourget Sport Climbing Venue      |                    1 |
| INV        | Invalides                            |                    0 |
| ELA        | Elancourt Hill                       |                    0 |
| STE        | Geoffroy-Guichard Stadium            |                    0 |
| LYO        | Lyon Stadium                         |                    0 |
| LGN        | Le Golf National                     |                    0 |
| PDP        | Parc des Princes                     |                    0 |
| LIL        | Pierre Mauroy Stadium                |                    0 |
| NIC        | Nice Stadium                         |                    0 |
| NAN        | La Beaujoire Stadium                 |                    0 |
```

Sample Answer for Question 4

5. Which athletes were born with the same data?
   ➔ Shows some interesting coincidences which can be used in promotion or news

```
SELECT DISTINCT A1.Athlete_Code,A1.Name,A1.Gender,A1.Birth_Date

FROM Athlete AS A1 JOIN Athlete AS A2 ON A1.Birth_Date = A2.Birth_Date AND

A1.Athlete_Code <> A2.Athlete_Code

ORDER BY A1.Birth_Date;
```

Figure 11-Query for Question 5

```
mysql> SELECT DISTINCT A1.Athlete_Code,A1.Name,A1.Gender,A1.Birth_Date
    -> FROM Athlete AS A1 JOIN Athlete AS A2 ON A1.Birth_Date = A2.Birth_Date AND
    -> A1.Athlete_Code <> A2.Athlete_Code
    -> ORDER BY A1.Birth_Date;
+--------------+-----------------------------+--------+------------+
| Athlete_Code | Name                        | Gender | Birth_Date |
+--------------+-----------------------------+--------+------------+
|      1535511 | RUIZ HURTADO Flor Denis     | Female | 1991-01-29 |
|      1537168 | CASTRO RIVERA Luis          | Male   | 1991-01-29 |
|      1537012 | SAIKHOM Mirabai Chanu       | Female | 1994-08-08 |
|      1535356 | MORENO Alexa                | Female | 1994-08-08 |
|      1538146 | CHAUHAN Maheshwari          | Female | 1996-07-04 |
|      1539952 | KENNEDY Terry               | Male   | 1996-07-04 |
|      1533364 | ISMAIL Mohamed              | Male   | 1997-01-01 |
|      1533371 | HASSAN Ibrahim              | Male   | 1997-01-01 |
|      1539008 | NOVAC Alexandru Mihaita     | Male   | 1997-03-24 |
|      1535364 | CARRILLO Duilio             | Male   | 1997-03-24 |
|      1538028 | BOSCO Eugenia               | Female | 1997-06-12 |
|      1537187 | FERNANDEZ GAMBOA Pedro Luis | Male   | 1997-06-12 |
|      1539548 | TIE Whitney                 | Female | 1999-04-07 |
|      1538622 | NURMATOV Shakhzod           | Male   | 1999-04-07 |
|      1537312 | REZAEI Mohammad Nabi        | Male   | 1999-04-10 |
|      1537286 | MARTINEZ Paulina            | Female | 1999-04-10 |
|      1536479 | VAZQUEZ Ana                 | Female | 2000-10-05 |
|      1538436 | AMINI POZVEH Fatemeh        | Female | 2000-10-05 |
|      1535199 | PORTILLO Erick              | Male   | 2000-10-05 |
|      1538632 | SAFAROLIYEV Sobirjon        | Male   | 2002-05-02 |
|      1538007 | CROOKS Jordan               | Male   | 2002-05-02 |
|      1539607 | SIDHU Vijayveer             | Male   | 2002-06-21 |
|      1535360 | TEJEDA Adirem               | Female | 2002-06-21 |
|      1535505 | LINARES Natalia             | Female | 2003-01-03 |
|      1538153 | SHKURDAI Anastasiya         | Female | 2003-01-03 |
+--------------+-----------------------------+--------+------------+
25 rows in set (0.02 sec)
```

Sample Answer for Question 5

6. What is the completed information of an official?
   ➔ Understand all the details about the official

```
Select Official.Official_Code, Official.Name, Gender, Belonged_Organization,
Event.Sport_Code,Event.Sport_Name
FROM Official NATURAL JOIN Event_Management NATURAL JOIN Event;
```

Figure 12-Query for Question 6

```
mysql> Select Official.Official_Code, Official.Name, Gender, Belonged_Organization,
    -> Event.Sport_Code,Event.Sport_Name
    -> FROM Official NATURAL JOIN Event_Management NATURAL JOIN Event;
+--------------+--------------------------------+--------+----------------------------------+------------+--------------------+
| Official_Code | Name                           | Gender | Belonged_Organization            | Sport_Code | Sport_Name         |
+--------------+--------------------------------+--------+----------------------------------+------------+--------------------+
|       1937701 | MESSER KERSTING Thomas         | Male   | Germany                          | BK3        | 3x3 Basketball     |
|       1937758 | BERTRAND Eric                  | Male   | Switzerland                      | BK3        | 3x3 Basketball     |
|       1937785 | HO Edmond                      | Male   | Hong Kong, China                 | BK3        | 3x3 Basketball     |
|       1937815 | MICHAELIDES Markos-Elias       | Male   | Switzerland                      | BK3        | 3x3 Basketball     |
|       1937848 | MALISZEWSKI Marek              | Male   | Poland                           | BK3        | 3x3 Basketball     |
|       1937858 | CHAJIDDINE Najib               | Male   | France                           | BK3        | 3x3 Basketball     |
|       1937879 | GHIZDAREANU Vlad               | Male   | Romania                          | BK3        | 3x3 Basketball     |
|       1937884 | JURAS Jasmina                  | Female | Serbia                           | BK3        | 3x3 Basketball     |
|       1937903 | SHI Qirong                     | Female | People's Republic of China       | BK3        | 3x3 Basketball     |
|       1937919 | KIM Gain                       | Female | Republic of Korea                | BK3        | 3x3 Basketball     |
|       1937973 | CSABAI-KASKOTO Brigitta        | Female | Hungary                          | BK3        | 3x3 Basketball     |
|       1937982 | OKATCH Dorothy                 | Female | Botswana                         | BK3        | 3x3 Basketball     |
|       1938013 | JACKSON Deanna                 | Female | United States of America         | BK3        | 3x3 Basketball     |
|       1938090 | TALISAINEN Alex                | Male   | Estonia                          | BK3        | 3x3 Basketball     |
|       1938099 | CSENDER Andrada Monika         | Female | Romania                          | BK3        | 3x3 Basketball     |
|       1938104 | VESKOVIC Vladimir              | Male   | Serbia                           | BK3        | 3x3 Basketball     |
|       1995222 | MEARY Aurelia                  | Female | France                           | BK3        | 3x3 Basketball     |
|       1995226 | BOIRE Elise                    | Female | France                           | BK3        | 3x3 Basketball     |
|       1995228 | BOUTONNET Gregoire             | Male   | France                           | BK3        | 3x3 Basketball     |
|       1995229 | PREL Nicolas                   | Male   | France                           | BK3        | 3x3 Basketball     |
|       1995231 | TAEYE Corinne                  | Female | France                           | BK3        | 3x3 Basketball     |
|       1995232 | FAYS Francois                  | Male   | France                           | BK3        | 3x3 Basketball     |
|       1995233 | FOREST Alexis                  | Male   | France                           | BK3        | 3x3 Basketball     |
|       1995236 | MECHIN Celine                  | Female | France                           | BK3        | 3x3 Basketball     |
|       1995237 | SEVAUX Arnaud                  | Male   | France                           | BK3        | 3x3 Basketball     |
|       1995239 | CORTINOVIS BEGLOT Chloe        | Female | France                           | BK3        | 3x3 Basketball     |
```

Sample Answer for Question 6

7. Who is the youngest athlete in the Olympic 2024?
   ➔ Show interesting fact that can be use in news

```
SELECT Athlete_Code, Name, Gender,ROUND((DATEDIFF(CURDATE(),Birth_Date) / 365),0) AS Age
FROM Athlete
WHERE (DATEDIFF(CURDATE(),Birth_Date) / 365)
<= ALL(SELECT DATEDIFF(CURDATE(),Birth_Date) / 365 FROM Athlete);
```

Figure 13-Query for Question 7

```
mysql> SELECT Athlete_Code, Name, Gender,ROUND((DATEDIFF(CURDATE(),Birth_Date) / 365),0) AS Age
    -> FROM Athlete
    -> WHERE (DATEDIFF(CURDATE(),Birth_Date) / 365)
    -> <= ALL(SELECT DATEDIFF(CURDATE(),Birth_Date) / 365 FROM Athlete);
+--------------+---------------------+--------+------+
| Athlete_Code | Name                | Gender | Age  |
+--------------+---------------------+--------+------+
|      1538415 | BEYRANVAND Mohammad | Male   |   16 |
+--------------+---------------------+--------+------+
```

Sample Answer for Question 7

8. Which events use most days to complete?
   ➔ The data obtained can be used for scheduling in future Olympic events

```
SELECT Sport_Name, MAX(DATEDIFF(Ended_Date, Started_Date)) AS 'Maximum Days used'
FROM Event NATURAL JOIN Participation
GROUP BY Sport_Name
HAVING MAX(DATEDIFF(Ended_Date, Started_Date)) =
(SELECT MAX(DATEDIFF(Ended_Date, Started_Date)) FROM Event NATURAL JOIN Participation);
```

Figure 14-Query for Question 8

```
mysql> SELECT Sport_Name, MAX(DATEDIFF(Ended_Date, Started_Date)) AS 'Maximum Days used'
    -> FROM Event NATURAL JOIN Participation
    -> GROUP BY Sport_Name
    -> HAVING MAX(DATEDIFF(Ended_Date, Started_Date)) =
    -> (SELECT MAX(DATEDIFF(Ended_Date, Started_Date)) FROM Event NATURAL JOIN Participation);
+------------+-------------------+
| Sport_Name | Maximum Days used |
+------------+-------------------+
| Football   |                17 |
| Handball   |                17 |
+------------+-------------------+
```

Sample Answer for Question 8

II.Advanced Features

➔The database has three stored procedure and two triggers

1) Stored Procedure

➔ All stored procedure implementing command are stored in StoredProcedure.sql script and then call by SettingUp.sql

Open ▾ | StoredProcedure.sql ~/Downloads/Assessment | Save

```
Delimiter $$ -- Set delimiter to $$ to create the stored procedure

DROP PROCEDURE IF EXISTS insAthlete$$
CREATE Procedure insAthlete(
    N varchar(50), -- Represent Athlete's Name
    G char(6), -- Represent Athlete's Gender
    D varchar(50), -- Represent Athlete's Discipline
    BD Date, -- Represent Athlete's Birth Date
    SJ varchar(50), -- Represent Athlete's Side Job
    CD char(3)) -- Represent Athlete's Country Code
Comment "This procedure insert a new Athlete in Athlete Table"
BEGIN
        Declare newAtCode int; -- newAtCode stored the new athlete code which is the latest athlete code the Athlete Table plus 1
        select Max(Athlete_Code)+1 from Athlete into newAtCode;
        IF (G IN ('Male','Female')) THEN -- Ensure Value of Gender is valid
            IF (SJ = '') THEN -- If athlete don't have side job then no value will be insert for Side_Job
                Insert into Athlete (Athlete_Code, Name, Gender, Discipline, Birth_Date, Side_Job, Country_Code) Values (newAtCode,N,G,D,BD,NULL,CD);
            ELSE
                Insert into Athlete (Athlete_Code, Name, Gender, Discipline, Birth_Date, Side_Job, Country_Code) Values (newAtCode,N,G,D,BD,SJ,CD);
            END IF;
        ELSE
            SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Gender of Athlete should be Male or Female!'; -- Output an error message to tell the user the inserted Gender is invalid
        END IF;
END$$

DROP PROCEDURE IF EXISTS insOfficial$$
CREATE Procedure insOfficial(
    N varchar(50), -- Represent Official's Name
    G char(6), -- Represent Official's Gender
    R varchar(30), -- Represent Official's Role
    BO varchar(80)) -- Represent Official's Belonged Organisation
Comment "This procedure insert a new Official in Official Table"
BEGIN
        Declare newOfCode int(7); -- newOfCode stored the new official code which is the latest official code the Official Table plus 1
        select Max(Official_Code)+1 from Official into newOfCode;
        IF (G IN ('Male','Female')) THEN -- Ensure Value of Gender is valid
            Insert into Official (Official_Code, Name, Gender, Role, Belonged_Organization) Values (newOfCode,N,G,R,BO);
        ELSE
            SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Gender of Official should be Male or Female!'; -- Output an error message to tell the user the inserted Gender is invalid
        END IF;
END$$

DROP PROCEDURE IF EXISTS showCountryMedalNumber$$
CREATE PROCEDURE showCountryMedalNumber()
Comment "This procedure display the total number of medal each country"
BEGIN
        Declare done int DEFAULT 0; -- Variable done acts as mark of cursor reach the end
        Declare countrycode CHAR(3); -- Represent Country Code
        -- Create a cursor that stored the code of country that have athlete who wins a medal
        Declare curMedal CURSOR FOR SELECT Country.Country_Code
                                    FROM (Country JOIN Athlete ON Country.Country_Code = Athlete.Country_Code)
                                                  JOIN Medal ON Athlete.Athlete_Code = Medal.Athlete_Code;
        Declare CONTINUE HANDLER FOR NOT FOUND Set done = 1; -- Set done = 1 if not more line is found in cursor (means cursor reach the end)

        Drop TABLE IF EXISTS Temporary;
        CREATE TEMPORARY TABLE Temporary(Country_Code CHAR(3),Name VARCHAR(40),TotalMedalNumber INT(100) DEFAULT 0);
        -- Temporary table stored the information of country and their total medal number accordingly
        -- This table with temporary value will be deleted when this procedure End

        INSERT INTO Temporary (Country_Code,Name) -- Insert the Country code and name into Temporary Table
        SELECT Country_Code,Name
        FROM Country;

        OPEN curMedal; -- Use the cursor created just now
        getCountMedal:LOOP -- The loop will fetch Country_Code in every line of cursor into the countrycode variable until it reach to the end of the cursor
        FETCH curMedal INTO countrycode;
            IF done = 1 THEN -- Means that cursor reach the end of data
                LEAVE getCountMedal; -- Exit from loop
            END IF;
            UPDATE Temporary SET TotalMedalNumber = TotalMedalNumber + 1 WHERE Country_Code = countrycode; -- TotalMedalNumber in Temporary plus 1 if the Country_Code in Temporary
table is same as countrycode in cursor
        END LOOP getCountMedal;
      CLOSE curMedal;

      Select * from Temporary ORDER BY TotalMedalNumber DESC; -- Display the content in Temporary table
END$$
```

Figure 15-Content in StoredProcedure.sql

➔ The stored procedure include:

i) insAthlete

➔ Insert new athlete in the Athlete table

```
mysql> call insAthlete('Yee','Male','Archery','2005-04-23','','IND');
Query OK, 1 row affected (0.01 sec)
```

Figure 16-Using insAthlete Stored Procedure

```
|      1540143 | FISCHER Guillermo                 | Male   | Handball                  | 1996-07-18 |
|      1540144 | Yee                               | Male   | Archery                   | 2005-04-23 |
+--------------+-----------------------------------+--------+---------------------------+------------+
401 rows in set (0.01 sec)
```

Advanced Feature Output 1

ii) insOfficial

➔ Insert new official in the Official table

```
mysql> call insOfficial('SecondYee','Male','Judge','HappyLand');
Query OK, 1 row affected (0.01 sec)
```

Figure 17-Using insOfficial Stored Procedure

```
|      4968543 | CAMPANILE Nicolas                 | Male   | Judge                     | International Gymnastics Federation |
|      4968544 | SecondYee                         | Male   | Judge                     | HappyLand                           |
+--------------+-----------------------------------+--------+---------------------------+-------------------------------------+
546 rows in set (0.00 sec)
```

Advanced Feature Output 2

iii) showCountryMedalNumber

➔ Output the countries and their total medals respectively

```
mysql> call showCountryMedalNumber();
+--------------+---------------------------------------+------------------+
| Country_Code | Name                                  | TotalMedalNumber |
+--------------+---------------------------------------+------------------+
| MEX          | Mexico                                |                3 |
| ETH          | Ethiopia                              |                2 |
| LCA          | Saint Lucia                           |                2 |
| IRL          | Ireland                               |                1 |
| JAM          | Jamaica                               |                1 |
| PUR          | Puerto Rico                           |                1 |
| AFG          | Afghanistan                           |                0 |
| AHO          | Netherlands Antilles                  |                0 |
| AIN          | AIN                                   |                0 |
```

Advanced Feature Output 3

2) Triggers

➔ All triggers implementing command are stored in Trigger.sql script and then call by SettingUp.sql

Trigger.sql
~/Downloads/Assessment

```
#YeeMingCheing_22467577

#Name:Yee Ming Cheing
#StudentID:22467577
#Database System 2024 Sem2
#Purpose:Create Trigger

Delimiter $$  -- Set delimiter to $$ to create the trigger

#Create trigger that ensure the new insert athlete age is not lower than minimum age which is 13
CREATE TRIGGER check_age_insert_athlete
    BEFORE INSERT ON Athlete -- trigger before any insert on Athlete table
    FOR EACH ROW
    BEGIN
        IF ROUND((DATEDIFF(CURDATE(),NEW.Birth_Date) / 365),0) < 13 THEN -- Means if age lower than 13
            SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Athlete must be at least 13 years old!'; -- Output an error message to tell the user the inserted Birth_Date is invalid
        END IF;
    END$$

#Create trigger that prevent wrong update on the value of Medal Type
CREATE TRIGGER change_medalcode_update_medal
    BEFORE UPDATE ON Medal -- trigger before any update on Medal table
    For Each ROW
    BEGIN
        IF NEW.Medal_Code = 1 AND NEW.Medal_Type <> 'Gold' THEN -- If code is 1 but type is not gold
            SET NEW.Medal_Type = 'Gold';
        ELSEIF NEW.Medal_Code = 2 AND NEW.Medal_Type <> 'Silver' THEN -- If code is 2 but type is not silver
            SET NEW.Medal_Type = 'Silver';
        ELSEIF NEW.Medal_Code = 3 AND NEW.Medal_Type <> 'Bronze' THEN -- If code is 3 but type is not bronze
            SET NEW.Medal_Type = 'Bronze';
        ELSEIF NEW.Medal_Code <> 1 OR NEW.Medal_Code <> 2 OR NEW.Medal_Code <> 3 THEN -- If code is not 1,2,or 3
            SIGNAL SQLSTATE '45000' SET MESSAGE_TEXT = 'Invalid Medal Code!'; -- Output an error message to tell the user the value of Medal_Code is invalid
        END IF;
    END$$

Delimiter ; -- Set the delimiter back to ; to prevent syntax error in mysql
```

Figure 18-Content in Trigger.sql

➔ The stored procedure include:

i) check_age_insert_athlete

➔ Check for the user inserted birth date's validity

```
mysql> call insAthlete('ThirdYee','Male','Archery','2024-04-23','','IND');
ERROR 1644 (45000): Athlete must be at least 13 years old!
mysql>
```

Figure 19-Evidence of Trigger 1 implementation

ii) change_medalcode_update_medal

➔ Ensure user updated medal's type follow the medal code

```
mysql> update Medal Set Medal_Code =3 where Athlete_Code=1539986;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
|      1537325 |           1 | Gold        | 2024-08-03 |
|      1537325 |           2 | Silver      | 2024-08-06 |
|      1539986 |           3 | Bronze      | 2024-08-03 |
+--------------+-------------+-------------+------------+
10 rows in set (0.00 sec)
```

Figure 20-Evidence of Trigger 2 implementation

III.Python implementation

The implementation of python program is related to three files:

i) Background.sql

➔A SQL script call by SettingUp.sql that set up the user's name and the password in MySQL to use the Python program

Open ▾ *Background.sql ~/Downloads/Assessment

```
#YeeMingCheing_22467577

#Name:Yee Ming Cheing
#StudentID:22467577
#Database System 2024 Sem2
#Purpose:Set up the User and the password

DROP USER 'Hello'@'localhost'; -- Drop the Hello User if exist

CREATE USER 'Hello'@'localhost' IDENTIFIED BY 'Pass@1234'; -- Create the Hello User with Password

GRANT ALL PRIVILEGES ON Olympic_22467577.* TO 'Hello'@'localhost'; -- Grant permissions to the Hello User for the database

FLUSH PRIVILEGES; -- Update the Changes
```

Figure 21-Content in Background.sql

ii) setting_env.sh

➔A bash scrip manually calls by user to set the environment variable USER and PASSWORD to create a secure connection between the Python program to database

Open ▾ *setting_env.sh ~/Downloads/Assessment

```
#!/bin/bash

if ! grep -q "export USER='Hello'" ~/.bashrc; then #Search "export USER='Hello'" in the file and if does not exist then
    echo "export USER='Hello'" >> ~/.bashrc #Set the Environment variable USER to Hello
fi #End If

if ! grep -q "export PASSWORD='Pass@1234'" ~/.bashrc; then #Search "export PASSWORD='Pass@1234'" in the file and if does not exist then
    echo "export PASSWORD='Pass@1234'" >> ~/.bashrc #Set the Environment variable PASSWORD USER to Hello
fi #End If

echo "Please enter 'source ~/.bashrc' in the shell to update the changes in the file"

#Way to run the script : ./setting_env.sh
```

Figure 22-Content in setting_env.sh

iii) ConnectOlympic.py

➔The Python program that connects to the Olympic_22467577 database which provide some useful functions such as delete or update athlete's information

Open ConnectOlympic.py ~/Downloads/Assessment

```
#YeeMingCheing_22467577

#Name:Yee Ming Cheing
#StudentID:22467577
#Database System 2024 Sem2
#Purpose:Python Connect to Database Olympic_22467577

import mysql.connector #Import mysql connector
import os #Import Operating System library (access environment variables)

#Get the password and user name from environment variables
DPASSWORD = os.getenv('PASSWORD')
DUSER = os.getenv('USER')

#Connect to Olympic_22467577 Database
db = mysql.connector.connect(
  host = "localhost", #hostname
  user = DUSER, #username
  passwd = DPASSWORD, #userpassword
  database = "Olympic_22467577") #databasename

#Create a cursor object to execute the queries
mycursor=db.cursor()

#Prepare the query
ShowAthlete = ("SELECT * FROM Athlete") #Display all the athlete
ShowWinner = ("SELECT * FROM Winner") #Display all the winner
InsertAthlete = ("CALL insAthlete(%s, %s, %s, %s, %s, %s)") #Insert an athlete using stored procedure
DeleteAthlete = ("DELETE FROM Athlete WHERE Athlete_Code = %s") #Delete specific athlete
UpdateAthlete = ("UPDATE Athlete SET Discipline =%s WHERE Athlete_Code = %s") #Update discipline of a specific athlete
#Display the code and the name of the athlete that has a side job and was born between 1995 and 2005
SampleQueryOne = ("SELECT Athlete_Code,Name,Side_Job FROM Athlete WHERE Side_Job IS NOT NULL AND Birth_Date BETWEEN '1995-01-01' AND '2005-12-31'")
#Display the code, the name, and the role of official come from a Sport Federation
SampleQueryTwo = ("SELECT Official_Code,Name,Role FROM Official WHERE Belonged_Organization LIKE '%Federation%'")

```

Figure 23-Section of content in ConnectOlympic.py

```
mysql> exit
Bye
yee@yee-virtual-machine:~/Downloads/Assessment$ python3 ConnectOlympic.py


Welcome
Please select one of the following option
1)Show Athlete Table
2)Show Winner Table
3)Insert a athlete into Athlete Table
4)Delete a athlete into Athlete Table
5)Update a athlete's disipline into Athlete Table
6)QueryOne(Display information of the athlete that has a side job and was born between 1995 and 2005)
7)QueryTwo(Display information of official from a Sport Federation)
0)Exit
Enter your choice: 6
(1533136, 'BASS BITTAYE Gina Mariam', 'police officer (sub-inspector)')
(1533176, 'CAMARA Ebrahima', 'prison officer')
(1533235, 'ARELLANO Fernanda', 'student')
(1533237, 'TOSCANO Pamela', 'sport soldier')
(1533364, 'ISMAIL Mohamed', 'sport soldier')
(1533371, 'HASSAN Ibrahim', 'soldier with the Djiboutian Republican Guard')
(1533372, 'WAISS Abdi', 'sport soldier')
(1533374, 'HASSAN NOUR Samiyah', 'sport soldier')
(1535191, 'GONZALEZ Alegna', 'serves in the navy')
(1535226, 'AGUILAR CHAVEZ PEON Mariana', 'student')
(1535310, 'BLANCO Luisa', 'student')
(1535343, 'WILLARS VALDEZ Randal', 'Armed forces  ')
(1535346, 'OROZCO LOZA Alejandra', 'Armed forces  ')
(1535348, 'AGUNDEZ GARCIA Gabriela', 'Armed forces  ')
(1535364, 'CARRILLO Duilio', 'Armed Forces  ')
```

Figure 24-Python Program successfully connect to the database

Discussion

A. Reflection

This is my first time creating a database system, and it took me 2 weeks to complete. In the beginning, I started with online research on the Olympics 2024 and got some useful information, including the data source. After that, I spent two days thinking of the ER model's draft and drawing it out. The draft helps me a lot in choosing the primary key and the data type of the attribute of each entity during the designing of the real database structure.

After creating the ER modal, I created the relational schema and implemented the database using MySQL. I found the database implementation quite straightforward, as I have done many lab exercises. In the following days, I created several queries, stored procedures, triggers, and a Python program. This process was interesting, as it clarified some of the concepts that I was struggling with during the lab session. Finally, I document the user guide and report, eventually completing my final assessment.

This assessment provides me with some valuable insights, such as the importance of having a draft before starting a project and the necessity of having a systematic plan to do the assessment.

B. Issues faced and solutions

Several challenges arose during the design and implementation of the database. Initially, I created 15 entities for my database and tried to implement them into the ER model. However, it was difficult for me to proceed with the next step; handling so many entities is more complex than I expected. Thus, I solve this problem by filtering the unnecessary entities, such as the schedule or event's category, and eventually reducing the total entities from 15 to 6, which allows me to effectively manage the ER Model and also the relational schema.

Another challenge was writing a Python program to connect to my database, as I had no experience in using Python. I discovered the syntax of Python is quite different from the programming languages I have learned, such as C++ or Java. To solve this, I visit some social media websites and watch tutorials on basic Python programming. I also researched ways to prevent hardcoding the user's name and password in the programming script. I spent 4 days repeatedly testing my Python program, and finally, it performed well.

Each of these challenges provided an opportunity for growth and a reminder for me to prepare well before doing the assessment in the future.

C. Future Directions

Due to the time limits, I was unable to cover or add some other features that I originally planned to my database system. During the implementation of the Python program, I was thinking about implementing a website that connects to my database. This website will provide a user-friendly interface for the users, especially those unfamiliar with Linux OS, to easily manage the Olympic data and use them in the research.

Besides, the second feature I want to add is the feedback system. The database will allow users to report bugs and also give suggestions. This feedback will be sent to my email every week and become a valuable reference for me to update the database system. With the help of the users, I believe that the database system will become more and more efficient and eventually become a well-known resource in sports research.

The last feature I hope to add is the import function. The user can import the excel file (.csv) to effectively add a large amount of data to the table. This feature not only increases the user's convenience but also improves its reusability.

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